

RUOLEI ZENG

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LLM RESEARCH AND EVALUATION

Computer Science Ph.D. researcher working on LLM context efficiency, answer verification, and adaptation. Published first-author work includes genomic language modeling; experience building reproducible GPU/Slurm evaluation workflows.

EDUCATION

University of Minnesota-Twin Cities | Ph.D., Computer Science

Expected May 2028 | Advisor: Shashi Shekhar

University of Wisconsin-Madison | B.S., Computer Science | Jun 2020

TECHNICAL SKILLS

ML: PyTorch, Transformers/PEFT, LoRA/IA3, BERT, GNNs, scikit-learn; distribution-shift evaluation, paired bootstrap.

Engineering: Python, SQL/SQLite, C++, Git, Linux, Slurm; multi-GPU/cloud experiments, mixed precision, checkpoint/resume, OOM-aware batching, reproducible evaluation.

RESEARCH EXPERIENCE

University of Minnesota | Spatial Computing Group

Ph.D. Researcher, Sep 2021-present | Research Assistant appointments since Jan 2024

- Automated multi-seed cloud GPU and UMN Slurm experiments with mixed precision, OOM-aware batching, caching, checkpoint/resume, retries, and failure logs.
- Built adaptation benchmarks over 218K soil-model records and 1M+ carbon-flux observations, testing geographic and environmental shifts with 1/5/10/20-shot settings and multiple seeds (ongoing).

SELECTED RESEARCH PROJECTS

LLM context selection with SABR

2026-present | Manuscript in preparation; targeting ICLR 2027 | Preliminary evaluation

- Developed SABR context selection for frozen LLMs using semantic retrieval, reranking, and structural packing under explicit token budgets.
- Evaluated Qwen2.5-7B/14B on 1,523 examples per model from BoolQ, SQuAD, HotpotQA, and single-span DROP, comparing nine methods/ablations across 30-90% token caps with paired bootstrap and selector-plus-generator cost accounting.
- At a 70% token cap on SQuAD (n=384), 14B SABR scored 78.4 F1 versus 75.7 for reranker-only selection, with 5.8% higher accounted amortized runtime.

LLM answer verification and inference cost

2026-present | Preliminary, unpublished

- Built a Qwen2.5-7B workflow to generate, audit, and revise GSM8K answers; evaluated error recovery and model-call cost.
- In a 300-question pilot, accuracy rose from 65.3% to 76.3% (+11.0 percentage points; paired bootstrap 95% CI: 6.0-16.0), using three calls versus one for the initial answer.
- Offline conditional-retry analysis reached 73.0% accuracy at 2.42 calls per question, versus 76.3% at three calls when always revising.

Language model tokenization and pretraining

Scientific Reports, 2025 [1] | Conceptualization and methodology | [Paper code](#)

- First author; co-led conceptualization and methodology for a task-oriented DNA dictionary, probabilistic segmentation, masked-language-model BERT pretraining, and multi-scale BERT-Inception classification.
- Evaluated 59,194 promoter sequences with 10-fold cross-validation; on non-TATA promoters, achieved 98.2% accuracy / 0.996 AUC, exceeding DNABERT by 1.2 percentage points in accuracy and 0.061 AUC.

SELECTED RESEARCH PROJECTS CONTINUED

Transfer learning with limited local labels

FTBSC-KGML: AAAI-26 AI2ASE Workshop [3] | RSC-GNN: ACM SIGSPATIAL 2026 [2]

- FTBSC-KGML: developed global-to-local fine-tuning, reducing validation MSE by 17.0-43.6% versus training only on the target state across Iowa, Illinois, and Indiana.
- RSC-GNN: added a target-specific residual network to a frozen source GNN; evaluated five held-out environmental subregions with 1-20% calibration labels across 218,367 hillslope nodes.
- RSC-GNN stored 5.2K target-specific parameters per region versus 14.5K for full fine-tuning (64% fewer). At 20% calibration, R^2 was 0.910 versus 0.901 without adaptation and 0.935 with full fine-tuning.

TailGuard for rare capability retention

2026-present | Ongoing research; targeting ACL 2027 | Preliminary synthetic study

- Evaluated targeted replay during Qwen2.5-32B training with 4-bit LoRA on a synthetic math/code/fact benchmark (3 training seeds, 3 generations); compared 5%/15% replay budgets with uniform replay, tracking rare-group, common-group, and overall accuracy.

Parameter-efficient adaptation

2026-present | Ongoing, unpublished

- Compared LoRA, risk-scaled LoRA, and IA3 on Qwen2.5-1.5B across ARC, CommonsenseQA, MMLU, MMLU-Pro, and TruthfulQA (3 seeds, 16-shot), analyzing negative transfer and leave-one-dataset-out performance.

Attention models for crop yield prediction

Agriculture, 2025 [4] | Conceptualization and software implementation

- First author; conceived the study and implemented Crossformer with local-perception and cross-window attention over 2,987 province-year records spanning three crops (1985-2022); used attention analysis and integrated gradients for interpretation.
- Reported R^2 of 0.981-0.986 across crops; on corn, reduced MSE by 73.4% versus a Transformer baseline.

Field image classification

WinterTurf | University of Minnesota | 2026

- Built a ConvNeXt-Tiny classification pipeline with duplicate and metadata checks; achieved 92.7% accuracy and 0.982 ROC-AUC on 164 held-out test images.

INDUSTRY EXPERIENCE

Predicine Corporation | Informatics Intern | Oct 2020-Aug 2021

- Built an SQLite-backed tool to retrieve genomic testing data, including database schema, indexing, and query design.

SELECTED PUBLICATIONS

First author: [1-4]. Coauthor: [5].

[1] **R. Zeng**, Z. Li, J. Li, and Q. Zhang. DNA promoter task-oriented dictionary mining and prediction model based on natural language technology. *Scientific Reports* 15, 153 (2025).

[2] **R. Zeng**, E. Luquin, D. Mulla, and S. Shekhar. Residual Site-Calibration of GNN Model for Spatially Heterogeneous Soil Organic Carbon Redistribution. *ACM SIGSPATIAL 2026*, main conference short paper, accepted.

[3] **R. Zeng** et al. Towards Fine-Tuning-Based Site Calibration for Knowledge-Guided Machine Learning: A Summary of Results. *AAAI-26 AI2ASE Workshop* (2026).

[4] **R. Zeng**, J. Li, Z. Li, and Q. Zhang. Crossformer-Based Model for Predicting and Interpreting Crop Yield Variations Under Diverse Climatic and Agricultural Conditions. *Agriculture* 15(9), 958 (2025).

[5] S. Zhang, A. Sharma, M. Farhadloo, M. Yang, **R. Zeng**, et al. Towards Surrogate Models with Hybrid Spatial Neural Networks: A Summary of Results. *ACM SIGSPATIAL GeoSIM Workshop*, pp. 57-69 (2025).

TEACHING AND SERVICE

Teaching Assistant, UMN (2021-2023; 2026): systems, discrete mathematics, databases, spatial computing.

Reviewer: KDD 2026 AI for Sciences track; Geoinformatica.